Concept of Operations
JoanSat

DPI Research

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This document has been prepared in accordance with ANSI/AIAA G-043A-2012, IEEE 1362-1998, and the requirements of 47 CFR Parts 25 and 97 of the Federal Communications Commission (FCC) regulations.

Revision History

Version	Date	Author	Description
1.0	May 02, 2026	Daniel Pang, Sophia Wang	Initial release

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1.0 Introduction

1.1 Purpose and Scope

This Concept of Operations (ConOps) document describes the operational concept for the JoanSat mission. It provides a comprehensive description of the mission objectives, system architecture, operational modes, communication architecture, and disposal plan. This document serves as the primary operational reference for FCC licensing under Part 25.114 (Standard Application) and supports frequency coordination with the IARU and ITU.

The ConOps is prepared in accordance with ANSI/AIAA G-043A-2012 (Guide for the Preparation of Operational Concept Documents) and IEEE 1362-1998.

1.2 Mission Objectives

- Fly the AmSat LTM board
- Perform Earth Imaging via COTS camera

1.3 Applicable Documents and Standards

Reference	Title
47 CFR 25.114	Applications for Space Station Authorizations
47 CFR 25.122	Streamlined Small Satellite Authorization Process
47 CFR 97.207	Space Station (Amateur Radio)
ANSI/AIAA G-043A-2012	Guide for Operational Concept Documents
IEEE 1362-1998	Concept of Operations Document Guide
CDS Rev 14.1	CubeSat Design Specification (Cal Poly)
NASA-STD-8719.14	Process for Limiting Orbital Debris
ITU Radio Regulations	International Frequency Coordination

1.4 Regulatory Framework

This mission will operate under FCC Part 25.114 (Standard Application). The application will include all required information per 47 CFR 25.114, including orbital parameters, frequency information, system operations narrative, interference analysis, and orbital debris mitigation plan.

The FCC requires compliance with orbital debris mitigation rules, including the 5-year post-mission deorbit requirement (FCC-22-74).

2.0 Mission Overview

2.1 Mission Description and Rationale

The JoanSat mission is a CubeSat designed to operate in Sun-Synchronous (SSO) at 500 km altitude. The primary payload is a Optical camera, LTM board for Ham comms, earth imaging. Operated by DPI Research, the mission targets a 1-year operational lifetime with full compliance to FCC regulations under Part 25.114 (Standard Application).

2.2 Stakeholders and Responsibilities

Stakeholder	Role	Responsibility
DPI Research	Mission Owner	Overall mission management and operations
	Launch Services	Launch vehicle integration and deployment
FCC	Regulatory Authority	Spectrum licensing and debris compliance
	Ground Segment	Communication and command operations

2.3 Mission Success Criteria

- Achieve stable orbit in 48 hours
- Successfully communicate with ground stations
- Sucessfully transmit ham communications
- Successfully image the earth
- Successfully download the images

2.4 Mission Constraints and Assumptions

Constraints: Mission lifetime shall not exceed 1 years. Spacecraft mass is 1 kg. All transmissions must be capable of immediate cessation via ground command.

Assumptions: Launch vehicle will deliver to target orbit within specified tolerances. Ground station availability meets minimum contact requirements. Solar power generation is sufficient for all operational modes.

2.5 Mission Timeline

Phase	Duration	Target Date
Design & Development	12-24 months	In progress
Integration & Test	3-6 months	TBD
FCC Filing & Coordination	6-12 months	TBD
Launch Readiness Review	1 month	TBD
Launch	—	
Nominal Operations	1 years	L + 1 years

Phase	Duration	Target Date
End-of-Life Disposal	< 5 years post-mission	Per FCC-22-74

3.0 Space Segment Description

3.1 Spacecraft Configuration

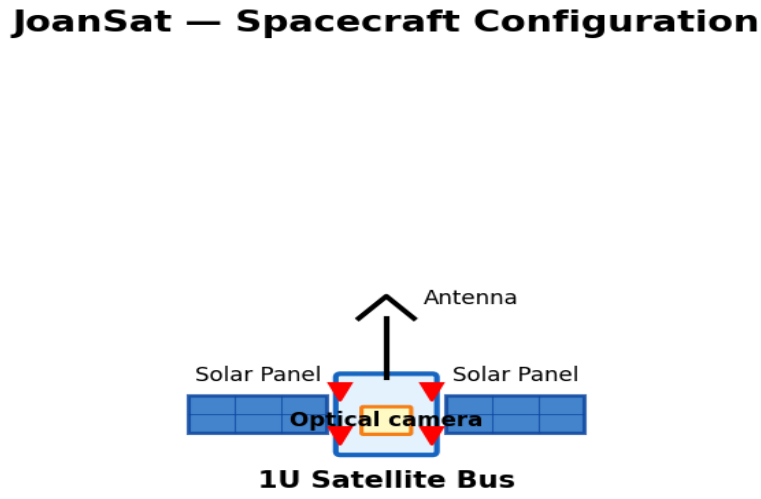

Figure 1: JoanSat spacecraft configuration diagram

Parameter	Value
Satellite Class	CubeSat
Form Factor	1U
Mass	1 kg
Dimensions	10 × 10 × 10 cm (1U)
Design Life	1 years
Attitude Control	
Propulsion	
Power System	
On-Board Computer	
Unique Telemetry Marker	Embedded in downlink beacon — per 47 CFR 25.122

Table 1: Spacecraft key parameters

3.2 Subsystem Summary

Subsystem	Description	Heritage / TRL
EPS		TRL 7-8
ADCS		TRL 7
C&DH		TRL 8
Comms	, bps	TRL 8
Thermal	Passive (MLI + surface coatings)	TRL 9
Propulsion		N/A
Payload	Optical camera, LTM board	TRL 5-6

3.3 Payload Description

The primary payload is a Optical camera, LTM board designed for Ham comms, earth imaging. Data is stored onboard and downlinked during scheduled ground station passes at bps.

4.0 Orbit Description

4.1 Orbital Parameters

Parameter	Value
Orbit Type	Sun-Synchronous (SSO)
Altitude	500 km
Inclination	50.4°
Eccentricity	~0 (near-circular)
RAAN	TBD
Orbital Period	94.5 minutes
Orbital Velocity	7.62 km/s
Orbits per Day	~15.2
LTAN	

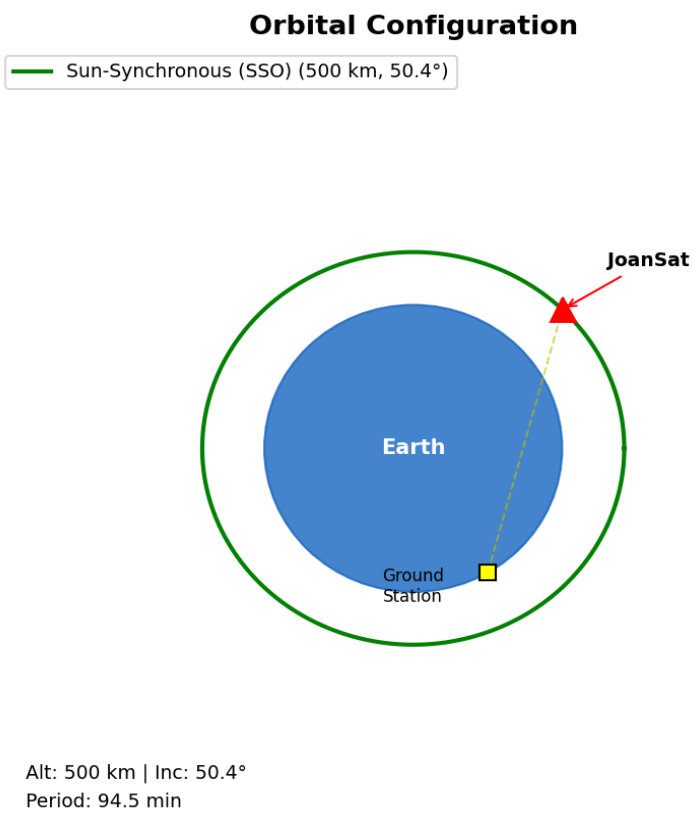

Figure 2: Orbital configuration diagram

4.2 Eclipse Analysis

At 500 km altitude, the spacecraft will experience approximately 33.1 minutes of eclipse per 94.5-minute orbit (~35% duty cycle). The power system is sized to sustain all essential subsystems through eclipse.

4.3 Orbital Decay Analysis

Estimated natural orbital decay time at 500 km: 3-7 years (natural decay, solar activity dependent). Active measures may be required to meet the FCC 5-year post-mission disposal requirement (FCC-22-74).

5.0 Mission Phases

Figure 3: Mission phase timeline

5.1 Pre-Launch (Integration & Testing)

Final spacecraft integration, environmental testing (vibration, thermal vacuum), end-to-end communication testing, and launch vehicle integration. FCC license application and IARU frequency coordination must be completed prior to launch.

5.2 Launch and Deployment

The spacecraft will be launched via . Post-deployment, the spacecraft will enter a pre-programmed inhibit period before activating any transmitters.

5.3 Early Operations / Detumbling

Upon deployment, the ADCS will execute detumbling to reduce tip-off rates. Initial beacon transmission will commence after the inhibit period. Ground station acquisition is expected within the first 24-48 hours.

5.4 Commissioning

Systematic checkout of all subsystems over approximately 2-4 weeks, including ADCS calibration, communication link verification, payload checkout, and flight software validation.

5.5 Nominal Operations

Primary mission operations lasting 1 years. The spacecraft will alternate between science data collection, data downlink, and housekeeping modes.

5.6 End-of-Life / Decommissioning

Upon mission completion, the spacecraft will enter disposal mode. All stored energy will be passivated. The spacecraft must deorbit within 5 years per FCC-22-74.

6.0 Operational Modes

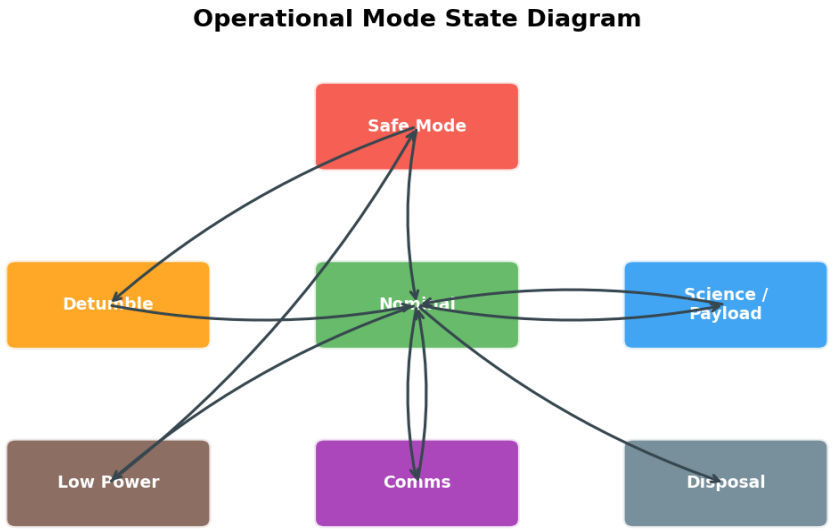

Figure 4: Operational mode state transition diagram

Mode	Description	Power (W)	Duration	ADCS State
Safe Mode	Minimum power, beacon only, sun-pointing	Low	Until recovery	Sun-pointing
Detumble	Reduce angular rates post-deployment	Low-Med	Min to hours	B-dot
Nominal	Standard housekeeping and attitude control	Medium	Default	3-axis
Science/Payload	Active payload data collection	High	Per schedule	Nadir/Target
Communication	Ground station pass, uplink/downlink	Med-High	~10 min/pass	GS tracking
Low Power	Battery conservation during eclipse	Low	Eclipse	Coarse
Disposal	Deorbit maneuver / passivation	Medium	One-time	As required

Table 2: Operational modes summary

6.1 Mode Transition Logic

Mode transitions are governed by the onboard flight software. Safe Mode is the highest priority and can be entered from any mode upon detection of an anomaly (under-voltage, communication timeout, attitude error exceedance). Ground command can override any mode transition.

7.0 Communication Architecture

Per 47 CFR 25.114(d)(1) and 25.122(a), this section provides the required narrative description of the communication system operations, frequencies, and spectrum-sharing strategy.

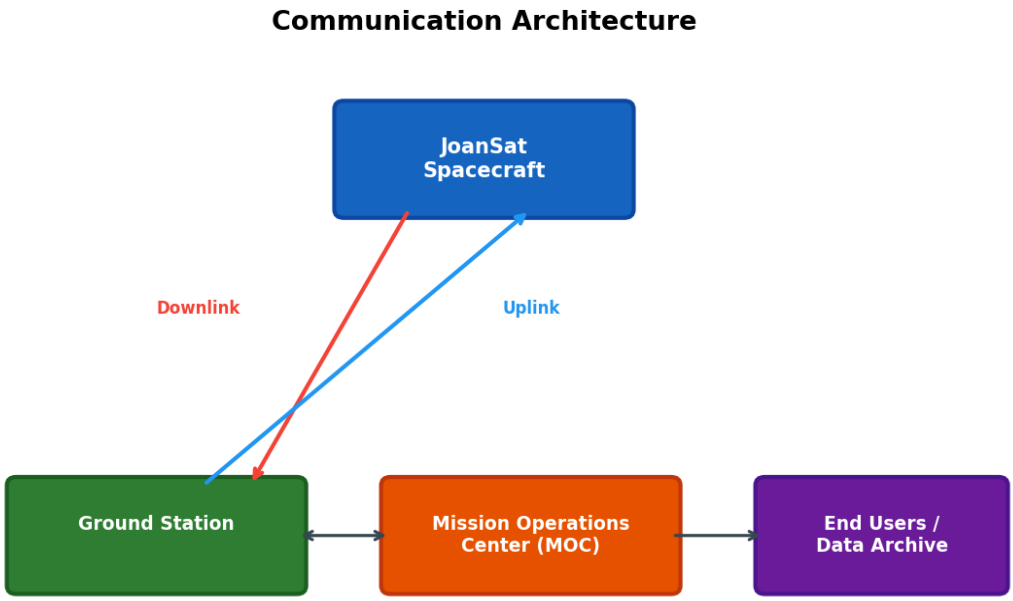

Figure 5: Communication architecture diagram

7.1 Frequency Plan

Parameter	Uplink	Downlink
Frequency		
Bandwidth		
Modulation	GMSK	GMSK
Data Rate	bps	bps
Protocol	AX.25	AX.25
Polarization	RHCP	RHCP
EIRP		—

7.2 Spectrum Sharing Strategy

The JoanSat mission is designed to share spectrum without constraining future satellite or terrestrial operators. The spacecraft uses narrowband transmissions with duty-cycled operation (active only during scheduled ground passes). IARU frequency coordination will be completed prior to launch.

7.3 Transmission Cessation Capability

Per 47 CFR 25.122(a) and 97.207, the spacecraft is capable of immediate cessation of all radio transmissions upon ground command. The telecommand link provides a dedicated 'TX OFF' command with highest priority in the command processing queue.

7.4 Link Budget Summary

Parameter	Uplink	Downlink
Tx Power		
Tx Antenna Gain		
Rx Antenna Gain		
System Noise Temp	1000 K	200 K
Required Eb/N0	10 dB	10 dB
Link Margin	> 6 dB	> 6 dB

Table 3: Link budget summary (see Appendix A for analysis)

8.0 Ground Segment

8.1 Ground Station Description

The primary ground station is located at . The station is equipped with a and . The ground station supports both manual and automated pass operations.

8.2 Contact Schedule

The ground station will have approximately 4 passes per day with an average duration of 10 minutes (total daily contact time: ~40 minutes).

8.3 Data Budget

Parameter	Value
Downlink Data Rate	9600 bps
Passes per Day	4
Daily Downlink Capacity	~2.7 MB
Daily Payload Data Generated	

9.0 Command and Data Handling

9.1 Command Protocol

Telecommands are transmitted via the uplink () using AX.25 protocol. Commands are authenticated to prevent unauthorized execution. Per 47 CFR 97.211, encrypted telecommand is permitted.

9.2 Telemetry and Health Monitoring

The spacecraft transmits periodic health beacons containing: battery voltage, solar panel current, temperatures, ADCS state, mode, and error flags. A unique signal-based telemetry marker (per 47 CFR 25.122) is embedded in every downlink frame.

9.3 Autonomy and Fault Management

The flight software implements a three-tier fault management architecture: (1) hardware watchdog timer, (2) software FDIR for subsystem anomalies, and (3) mission-level safe mode entry upon detection of critical faults.

10.0 Power Budget Analysis

This section presents the power generation and consumption analysis for each operational mode.

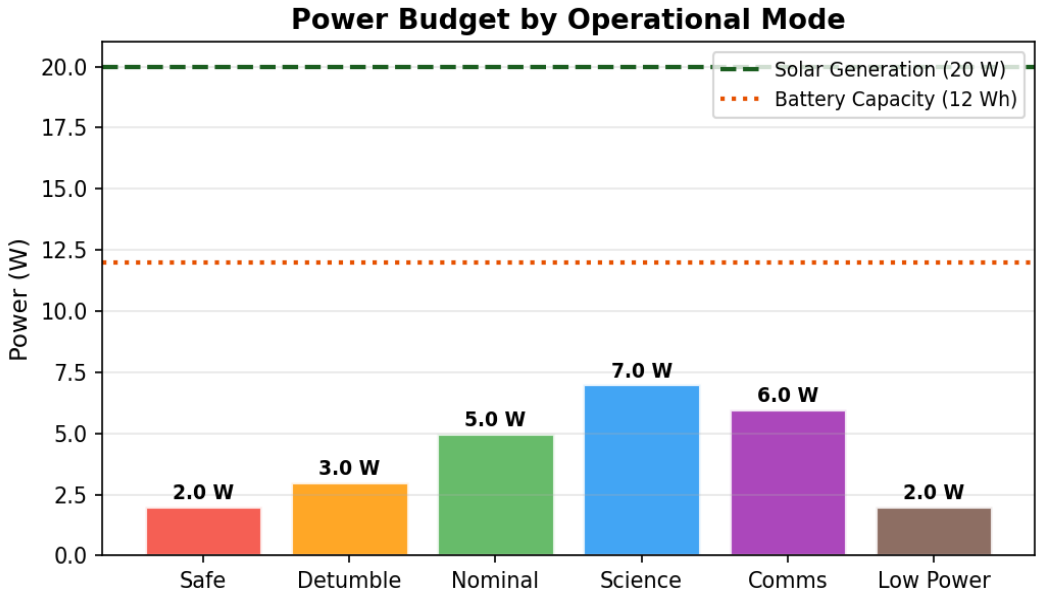

Figure 6: Power budget by operational mode

Parameter	Value
Peak Solar Generation	20 W
Battery Capacity	12 Wh
Battery Type	Li-ion
Solar Panel Area	
Eclipse Duration (per orbit)	33.1 min

11.0 Orbital Debris Mitigation Plan

Per 47 CFR 25.114(d)(14) and FCC-22-74 (5-year deorbit rule).

11.1 Operational Debris Release

No debris will be released during normal mission operations. All deployment mechanisms use captive fasteners.

11.2 Collision Risk Assessment

Requirement	Threshold	Assessment	Compliance
Small debris collision (<10 cm)	P < 0.01	TBD (NASA DAS)	Expected compliant
Large object collision (>10 cm)	P < 0.001	TBD (NASA DAS)	Expected compliant
Accidental explosion	Minimize probability	Passivation at EOL	Compliant

11.3 Trackability

The JoanSat spacecraft meets minimum dimension requirements for tracking by the 18th Space Defense Squadron Space Surveillance Network. The spacecraft will be registered with the U.S. Space Command catalog.

11.4 Conjunction Assessment and Response

The mission operations team will subscribe to conjunction data messages (CDMs) from the 18th Space Defense Squadron. Ephemeris data will be shared with the Space Surveillance Network to support conjunction screening.

12.0 End-of-Life Disposal

Per 47 CFR 25.114(d)(14) and FCC-22-74 (5-year deorbit rule).

12.1 Deorbit Method

Disposal method: Atmospheric reentry (uncontrolled) — natural drag.

Parameter	Value
Disposal Method	Atmospheric reentry (uncontrolled) — natural drag
Post-Mission Orbital Lifetime	< 5 years (per FCC-22-74)
Disposal Success Probability	≥ 0.9 (per 47 CFR 25.122)
Human Casualty Risk	< 1:10,000 (per 47 CFR 25.114)

12.2 Passivation

At end of life, all stored energy will be removed: batteries discharged, pressurized systems vented, reaction wheels de-spun. This eliminates accidental explosion risk per 47 CFR 25.114(d)(14)(ii).

12.3 Casualty Risk Assessment

For a 1 kg satellite, complete demise during atmospheric reentry is expected, resulting in zero casualty risk.

13.0 Risk Assessment and Contingency Operations

13.1 Mission Risk Register

ID	Risk	Likelihood	Impact	Mitigation
R1	Launch delay	Medium	Medium	Flexible manifest, rideshare options
R2	Communication loss	Low	High	Redundant GS, autonomous safe mode
R3	ADCS failure	Low	High	Backup modes, sun-pointing safe mode
R4	Power degradation	Medium	Medium	Oversized arrays, battery management
R5	Payload malfunction	Low	Medium	Pre-flight testing, calibration
R6	Debris collision	Very Low	Very High	Conjunction monitoring, avoidance
R7	Software anomaly	Medium	Medium	Watchdog, safe mode, patching
R8	Deorbit failure	Low	High	Drag augmentation backup

Table 5: Mission risk register

13.2 Contingency Operations

Loss of Communication: If no ground contact within 72 hours, spacecraft will autonomously cycle through communication modes. After 30 days, permanent safe mode.

Safe Mode Recovery: Entered upon under-voltage, attitude error, or comm timeout. Recovery requires ground command after root cause analysis.

14.0 Regulatory Compliance Summary

Requirement	Regulation	Status	Section
FCC Space Station License	Part 25.114 (Standard Application)	In preparation	§1.4
Orbital Parameters	47 CFR 25.114(c)	Provided	§4.0
System Operations Narrative	47 CFR 25.114(d)(1)	Provided	§5.0–§9.0
Communication Architecture	47 CFR 25.114(c)(4)	Provided	§7.0
Spectrum Sharing Strategy	47 CFR 25.122(a)	Provided	§7.2
Transmission Cessation	47 CFR 25.122(a) / 97.207	Implemented	§7.3
Unique Telemetry Marker	47 CFR 25.122(a)	Designed	§3.1
Debris Mitigation Plan	47 CFR 25.114(d)(14)	Provided	§11.0
Collision Risk Assessment	47 CFR 25.114(d)(14)(iii)	Pending	§11.2
Trackability	47 CFR 25.114(d)(14)(v)	Compliant	§11.3
Conjunction Response	47 CFR 25.114(d)(14)(vi)	Planned	§11.4
5-Year Deorbit	FCC-22-74	Compliant	§12.0
Casualty Risk < 1:10,000	47 CFR 25.114(d)(14)(iv)	Assessed	§12.3
Passivation Plan	47 CFR 25.114(d)(14)(ii)	Planned	§12.2
IARU Frequency Coordination	IARU Form V40	In preparation	§7.1

Table 6: FCC regulatory compliance cross-reference matrix

15.0 References

- 47 CFR Part 25 — Satellite Communications (https://www.law.cornell.edu/cfr/text/47/part-25)
- 47 CFR 25.114 — Applications for Space Station Authorizations (https://www.law.cornell.edu/cfr/text/47/25.114)
- 47 CFR 25.122 — Streamlined Small Satellite Process (https://www.law.cornell.edu/cfr/text/47/25.122)
- 47 CFR Part 97 — Amateur Radio Service (https://www.law.cornell.edu/cfr/text/47/part-97)
- 47 CFR 97.207 — Space Station (https://www.law.cornell.edu/cfr/text/47/97.207)
- ANSI/AIAA G-043A-2012: Guide for the Preparation of Operational Concept Documents
- IEEE 1362-1998: IEEE Guide for Information Technology — System Definition — Concept of Operations Document
- CubeSat Design Specification Rev 14.1 (https://www.cubesat.org/cubesatinfo)
- NASA CubeSat Launch Initiative Resources (https://www.nasa.gov/content/cubesat-launch-initiative-resources)
- NASA GSFC-STD-1001A: Flight Project Lifecycle Reviews (https://standards.nasa.gov/standard/GSFC/GSFC-STD-1001)
- IARU Satellite Frequency Coordination (https://www.iaru.org/reference/satellites/)
- FCC 5-Year Deorbit Rule, FCC-22-74 (2022)
- ITU Radio Regulations (https://www.itu.int/pub/R-REG-RR)

Appendix A: Link Budget Analysis

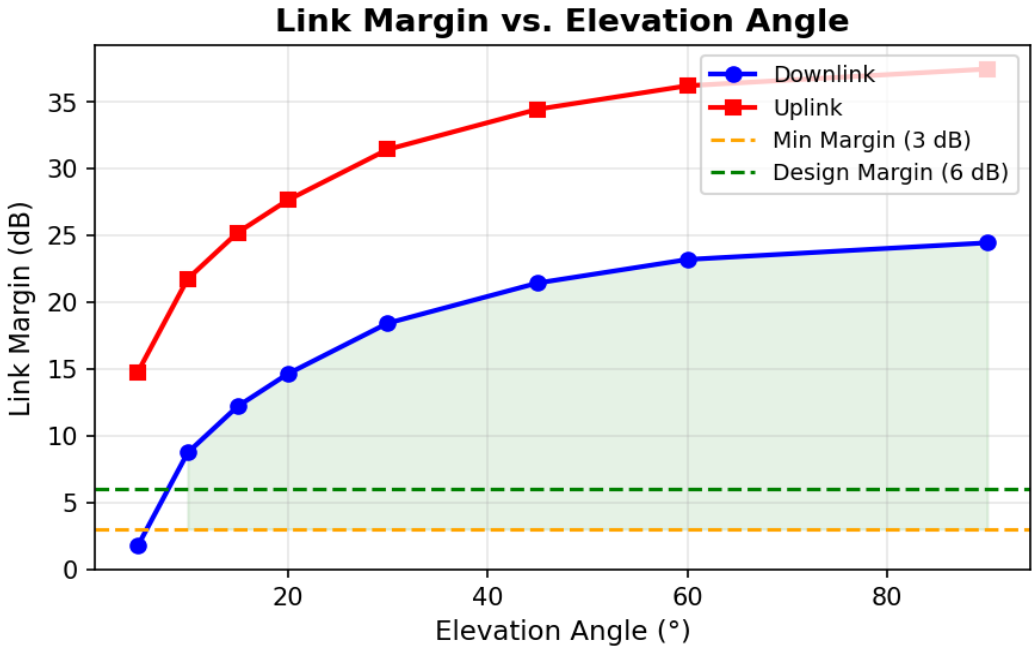

Figure 7: Link margin vs. elevation angle

Parameter	5° Elevation (Worst)	90° Elevation (Best)
Slant Range	5737 km	500 km
Free Space Path Loss	160.4 dB	139.2 dB
Atmospheric Loss	2.0 dB	0.5 dB
Total Path Loss	163.4 dB	140.2 dB

Appendix B: FCC Application Cross-Reference Matrix

FCC Requirement	CFR Citation	ConOps Section
Applicant Information	25.114(a)	Title Page
Orbital Parameters	25.114(c)(1)-(5)	§4.0
Frequency Information	25.114(c)(4)	§7.1
Antenna Characteristics	25.114(c)(4)(iii)	§7.4
System Operations	25.114(d)(1)	§5.0–§9.0
Interference Analysis	25.114(d)(3)	§7.2
Cessation of Emissions	25.122(a)	§7.3
Debris Mitigation	25.114(d)(14)	§11.0
Collision Avoidance	25.114(d)(14)(iii)	§11.2, §11.4
End-of-Life Disposal	25.114(d)(14)(iv)	§12.0
Trackability	25.114(d)(14)(v)	§11.3
Casualty Risk	25.114(d)(14)(iv)	§12.3
Unique Telemetry Marker	25.122(a)	§3.1
Maneuverability	25.122(a)	§3.1

Appendix C: Acronyms and Abbreviations

Acronym	Definition
ADCS	Attitude Determination and Control System
C&DH	Command and Data Handling
CDM	Conjunction Data Message
ConOps	Concept of Operations
DAS	Debris Assessment Software
EIRP	Effective Isotropic Radiated Power
EOL	End of Life
EPS	Electrical Power System
FCC	Federal Communications Commission
FDIR	Fault Detection, Isolation, and Recovery
GEO	Geostationary Earth Orbit
IARU	International Amateur Radio Union
ITU	International Telecommunication Union
LEO	Low Earth Orbit
LTAN	Local Time of Ascending Node
MEO	Medium Earth Orbit
MLI	Multi-Layer Insulation
MOC	Mission Operations Center
NGSO	Non-Geostationary Orbit
RAAN	Right Ascension of the Ascending Node
RHCP	Right-Hand Circular Polarization
SDS	Space Defense Squadron
SSO	Sun-Synchronous Orbit
TRL	Technology Readiness Level